

# Basic Properties and Contraction of AFEM

## Adaptive Finite Element Methods for Elliptic PDEs — Lecture 7

Pedro Morin    Ricardo H. Nochetto



[github.com/morinpedro/afem-cimpa-Rosario-2026](https://github.com/morinpedro/afem-cimpa-Rosario-2026)

# Convergence of SOLVE $\rightarrow$ ESTIMATE $\rightarrow$ MARK $\rightarrow$ REFINE

We consider two scenarios

- **Piecewise constant data and interior node property:** We assume that
  - The coefficient  $A(x)$  and the forcing term  $f(x)$  are constant on each element of the coarsest initial mesh  $\mathcal{T}_0$ .
  - Each marked element is refined in such a way that an interior node is created inside it.

$\rightsquigarrow$  Contraction of the error  $\|u - U_k\|$ .

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$$\rightsquigarrow \quad \text{Contraction of the error } \|u - U_k\|.$$

- **General case:** We assume that  $A(x)$  and  $f(x)$  are piecewise smooth and do not enforce the interior node property. We prove

- A *localized* upper bound and for  $\|U_k - U_{k+1}\|$
- How a single refinement changes  $U_k$  and the estimator  $\mathcal{E}_k$  is thus reduced.

$$\rightsquigarrow \quad \text{Contraction of the } \textit{quasi error} \quad \zeta_k = \left( \|u - U_k\|^2 + \gamma \mathcal{E}_k^2 \right)^2.$$

# Roadmap

- 1 The adaptive loop: notation
- 2 Four basic properties
- 3 Contraction: vanishing oscillation
- 4 Estimator reduction
- 5 Contraction: general data
- 6 Synthesis
- 7 Localized upper bound

The adaptive loop: notation

## Nested meshes from AFEM

- **Nested meshes:** The loop of Lectures 5–6 produces a sequence of nested conforming meshes

$$\mathcal{T}_0 \leq \mathcal{T}_1 \leq \mathcal{T}_2 \leq \cdots, \quad \mathbb{V}_k := \mathbb{V}_{\mathcal{T}_k} \subset \mathbb{V}_{k+1},$$

with Galerkin solutions  $U_k \in \mathbb{V}_k$ , indicators  $\mathcal{E}_k(U_k, T)$ , and marked sets  $\mathcal{M}_k \subset \mathcal{T}_k$  chosen by Dörfler marking.

- **Nestedness is automatic:** bisection never coarsens, so  $\mathbb{V}_k \subset \mathbb{V}_{k+1}$  always.
- **Central object:** The single-step increment  $U_{k+1} - U_k \in \mathbb{V}_{k+1}$  is central below: it appears in both the localized upper bound and in the estimator reduction.

### Abbreviations used throughout

$$e_k = \|u - U_k\|_{\Omega}, \quad E_k = \|U_{k+1} - U_k\|_{\Omega}, \quad \mathcal{E}_k = \mathcal{E}_k(U_k, \mathcal{T}_k), \quad \text{osc}_k = \text{osc}_k(U_k).$$

Four basic properties

# Four basic properties

Specialized to nested meshes  $\mathcal{T}_k \leq \mathcal{T}_{k+1}$ :

- ❶ **Upper bound** (reliability, Lecture 5):  $e_k^2 \leq C_1 \mathcal{E}_k^2$ .
- ❷ **Lower bound** (efficiency, Lecture 5):  $C_2 \mathcal{E}_k^2 \leq e_k^2 + \text{osc}_k^2$ .

- ❸ **Discrete lower bound** (thanks to interior node property, Lecture 5):

$$C_2 \mathcal{E}_k^2(U_k, T) \leq \|U_{k+1} - U_k\|_{\omega_k(T)}^2 + \text{osc}_k^2(U_k, \omega_k(T)).$$

- ❹ **Pythagoras orthogonality:**  $e_k^2 = e_{k+1}^2 + E_k^2$ ,

because  $\mathcal{B}[u - U_{k+1}, U_{k+1} - U_k] = 0$  for  $U_{k+1} - U_k \in \mathbb{V}_{k+1}$ .



Contraction: vanishing oscillation

## The easy case: no oscillation

- **Piecewise constant data:** When  $f, A \in \mathcal{P}_0(\mathcal{T}_0)$  (piecewise constant on the *initial* mesh),  $\text{osc}_k(U_k) \equiv 0$  on every refinement, and the estimator is equivalent to the energy error with **no correction**.
- **Direct contraction:** We can then contract  $e_k$  *directly*, converting Dörfler marking into an energy decrement via the interior node property.

# Contraction theorem (no oscillation)

## Theorem 1 (contraction, piecewise constant data)

Let  $\text{osc}_k(U_k) = 0$  and  $\mathcal{M}_k$  satisfy Dörfler marking  $\mathcal{E}_k(U_k, \mathcal{M}_k) \geq \theta \mathcal{E}_k(U_k, \mathcal{T}_k)$ ,  $0 < \theta < 1$ . If every  $T \in \mathcal{M}_k$  gets an interior node of  $\mathcal{T}_{k+1}$ , then

$$e_{k+1} \leq \alpha e_k, \quad \alpha = \left(1 - \theta^2 \frac{C_2}{C_1}\right)^{1/2} < 1.$$

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$$e_{k+1} \leq \alpha e_k, \quad \alpha = \left(1 - \theta^2 \frac{C_2}{C_1}\right)^{1/2} < 1.$$

- **Proof:** By Pythagoras,  $e_{k+1}^2 = e_k^2 - E_k^2$ ; bound  $E_k^2$  below via the interior node property, Dörfler marking, and the upper bound, in this order:

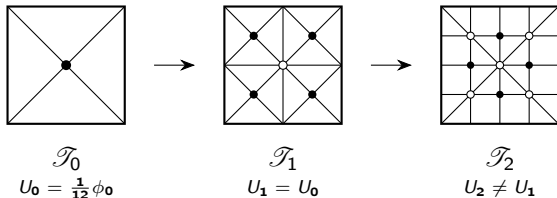
$$E_k^2 \geq C_2 \mathcal{E}_k^2(U_k, \mathcal{M}_k) \geq \theta^2 C_2 \mathcal{E}_k^2 \geq \theta^2 \frac{C_2}{C_1} e_k^2. \quad \square$$

- **Linear convergence:** Iterating: **linear convergence**  $e_k \leq \alpha^k e_0 \rightarrow 0$ .

# Why the interior node property is necessary

## Counterexample (Morin–Nochetto–Siebert)

- $A = I$ ,  $f = 1$ ,  $\Omega = (0, 1)^2$ , criss-cross initial mesh  $\mathcal{T}_0$ .
- Two bisection steps adding *no* interior node leaves the Galerkin solution *unchanged*:  $U_1 = U_0 = \frac{1}{12}\phi_0$ , so
- $E_0 = 0$  and  $e_0 = e_1$  — **no contraction**.
- Only  $\mathcal{T}_2$ , which finally introduces an interior node, gives  $U_2 \neq U_1$ .



Estimator reduction

## Setting: the estimator across a refinement step

### Estimator recap (Lecture 5):

$$\mathcal{E}_{\mathcal{T}}(U) = \left( \sum_{T \in \mathcal{T}} \mathcal{E}_{\mathcal{T}}^2(U, T) \right)^{1/2}, \quad \mathcal{E}_{\mathcal{T}}^2(U, T) = h_T^2 \|r\|_{L^2(T)}^2 + h_T \|j\|_{L^2(\partial T)}^2.$$

### Two competing effects between $\mathcal{T}_k$ and $\mathcal{T}_{k+1}$

- New elements are **smaller**: this shrinks the  $h$ -weighted indicators on refined elements.
- The discrete solution changes from  $U_k$  to  $U_{k+1}$ : this **perturbs** every indicator, but only in a controlled, Lipschitz way.

- **Plan:** We quantify each effect, then combine them.

# Lipschitz dependence on the discrete argument

## Lemma 1 (Lipschitz dependence)

For  $W_1, W_2 \in \mathbb{V}_{\mathcal{T}}(\mathcal{T})$  and each  $T \in \mathcal{T}$ ,

$$|\mathcal{E}_{\mathcal{T}}(W_1, T) - \mathcal{E}_{\mathcal{T}}(W_2, T)| \lesssim \|\nabla(W_1 - W_2)\|_{L^2(\omega_T)}.$$

Therefore  $|\mathcal{E}_{\mathcal{T}}(W_1) - \mathcal{E}_{\mathcal{T}}(W_2)| \leq \sqrt{C_0} \|W_1 - W_2\|_{\Omega}.$

**Sketch of proof:** Recall that  $\mathcal{E}_{\mathcal{T}}(W, T)^2 = h_T^2 \|\operatorname{div}(A \nabla W) + f\|_{L^2(T)}^2 + h_T \|\llbracket A \nabla W \rrbracket\|_{L^2(\partial T)}^2$

Let  $W_1, W_2 \in \mathbb{V}_{\mathcal{T}}(\mathcal{T})$ , and  $T \in \mathcal{T}$ . For the interior residual we proceed as follows:

$$\begin{aligned} \left| h_T \|f + \operatorname{div}(A \nabla W_1)\|_{L^2(T)} - h_T \|f + \operatorname{div}(A \nabla W_2)\|_{L^2(T)} \right| &\leq h_T \|\operatorname{div}(A \nabla W_1) - \operatorname{div}(A \nabla W_2)\|_{L^2(T)} \\ &= h_T \|\operatorname{div}(A \nabla(W_1 - W_2))\|_{L^2(T)}. \end{aligned}$$

Let  $W = W_1 - W_2$

$$\begin{aligned} h_T \|\operatorname{div}(A \nabla(W))\|_{L^2(T)} &\leq h_T \|\operatorname{div}(A)\|_{L^\infty(T)} \|\nabla W\|_{L^2(T)} + h_T \|A\|_{L^\infty(T)} \|D^2 W\|_{L^2(T)} \\ &\lesssim h_T \|\nabla W\|_{L^2(T)} + h_T h_T^{-1} \|\nabla W\|_{L^2(T)} \lesssim \|\nabla W\|_{L^2(T)} = \|\nabla(W_1 - W_2)\|_{L^2(T)}. \end{aligned}$$



# Estimator reduction: statement

## Proposition 1 (estimator reduction)

Let  $\mathcal{T}_{k+1} \geq \mathcal{T}_k$  bisect every marked  $T \in \mathcal{M}_k$  at least once, and set  $\lambda := 1 - 2^{-1/d} \in (0, 1)$ . There is  $C_0 > 0$  (shape regularity and data only) such that for every  $\delta > 0$ ,

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta) \left[ \mathcal{E}_k^2(U_k) - \lambda \mathcal{E}_k^2(U_k, \mathcal{M}_k) \right] + (1 + \delta^{-1}) C_0 E_k^2.$$

## Reading it

The bracket says: refining  $\mathcal{M}_k$  *shrinks* the estimator by a fixed fraction  $\lambda$  of its marked part. The last term is the price of the solution having changed — and it is controlled, via the localized upper bound, by the very same marked set.

## Proof — Step 1: reduction for a *fixed* function

- Fixed function:** Fix  $V \in \mathbb{V}_k \subset \mathbb{V}_{k+1}$ . Bisecting  $T$  once halves  $|T|$ , so  $h_{T'} \leq 2^{-1/d} h_T$  for a descendant  $T' \subset T$ ; since  $V$  stays polynomial on descendants and interior edges created *inside*  $T$  carry no jump,

$$\sum_{T' \subset T} \mathcal{E}_{k+1}^2(V, T') \leq 2^{-1/d} \mathcal{E}_k^2(V, T).$$

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$$\sum_{T' \subset T} \mathcal{E}_{k+1}^2(V, T') \leq 2^{-1/d} \mathcal{E}_k^2(V, T).$$

- Summation:** Unrefined elements keep their indicator unchanged. Summing over  $\mathcal{T}_k$  and isolating the marked (hence refined) elements  $\mathcal{M}_k \subset \mathcal{R}_k = \mathcal{T}_k \setminus \mathcal{T}_{k+1}$ :

$$\mathcal{E}_{k+1}^2(V) \leq \underbrace{\mathcal{E}_k^2(V) - (1 - 2^{-1/d})}_{\lambda} \sum_{T \in \mathcal{R}_k} \mathcal{E}_k^2(V, T) \leq \mathcal{E}_k^2(V) - \lambda \mathcal{E}_k^2(V, \mathcal{M}_k).$$

## Proof — Step 2: combine with the Lipschitz bound

- **Young's inequality:**  $(a + b)^2 \leq (1 + \delta)a^2 + (1 + \delta^{-1})b^2$ , applied to

$$\mathcal{E}_{k+1}(U_{k+1}) = \mathcal{E}_{k+1}(U_k) + (\mathcal{E}_{k+1}(U_{k+1}) - \mathcal{E}_{k+1}(U_k)):$$

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (\mathcal{E}_{k+1}(U_k) + \sqrt{C_0} E_k)^2 \leq (1 + \delta) \mathcal{E}_{k+1}^2(U_k) + (1 + \delta^{-1}) C_0 E_k^2,$$

using the Lipschitz Lemma with  $W_1 = U_{k+1}$ ,  $W_2 = U_k$  on  $\mathcal{T}_{k+1}$ .

- **Apply Step 1:** Now apply Step 1 with  $V = U_k$  to  $\mathcal{E}_{k+1}^2(U_k)$ :

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta) \left[ \mathcal{E}_k^2(U_k) - \lambda \mathcal{E}_k^2(U_k, \mathcal{M}_k) \right] + (1 + \delta^{-1}) C_0 E_k^2. \quad \square$$

Estimator reduction is the engine of the *general* contraction proof, next.

Contraction: general data

# Estimator reduction: review

## Proposition 2 (estimator reduction)

Let  $\mathcal{T}_{k+1} \geq \mathcal{T}_k$  bisect every marked  $T \in \mathcal{M}_k$  at least once, and set  $\lambda := 1 - 2^{-1/d} \in (0, 1)$ . There is  $C_0 > 0$  (shape regularity and data only) such that for every  $\delta > 0$ ,

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta) \left[ \mathcal{E}_k^2(U_k) - \lambda \mathcal{E}_k^2(U_k, \mathcal{M}_k) \right] + (1 + \delta^{-1}) C_0 E_k^2.$$

If we combine with Dörfler's Marking Strategy

$$\mathcal{E}_k^2(U_k, \mathcal{M}_k) \geq \theta \mathcal{E}_k^2(U_k) \quad \implies \quad \mathcal{E}_k^2(U_k) - \lambda \mathcal{E}_k^2(U_k, \mathcal{M}_k) \leq \mathcal{E}_k^2(U_k) - \lambda \theta^2 \mathcal{E}_k^2(U_k).$$

Then

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta)(1 - \lambda \theta^2) \mathcal{E}_k^2(U_k) + (1 + \delta^{-1}) C_0 E_k^2.$$

## Why energy alone can stall

- **Energy stalls:** In the counterexample,  $U_{k+1} = U_k$  so  $E_k = 0$  and  $e_{k+1} = e_k$ : the energy error **stalls**.
- **Estimator still drops:** Yet with  $E_k = 0$ , estimator reduction gives

$$\mathcal{E}_{k+1}^2(U_{k+1}) \leq (1 + \delta)(1 - \lambda\theta^2) \mathcal{E}_k^2(U_k),$$

which *strictly decreases* for small  $\delta > 0$ .

- **Neither alone works:** Neither  $e_k$  nor  $\mathcal{E}_k(U_k)$  contracts *alone* in general — energy can stall while the estimator drops, and the estimator is not even monotone on its own (it depends on the changing  $U_k$ ).

# The quasi-error

The fix: a weighted combination

$$\zeta_k^2 := e_k^2 + \gamma \mathcal{E}_k^2 \quad (\gamma > 0 \text{ to be chosen}).$$

- **Blend of two worlds:** This blends a *continuous* quantity ( $e_k$ ) with a *discrete* one ( $\mathcal{E}_k$ ), neither of which suffices alone — the hallmark of the Cascón–Kreuzer–Nochetto–Siebert convergence theory.
- **Goal:** We show  $\zeta_k$  contracts, for *any* data, with or without oscillation, with or without the interior node property.



# Contraction of the quasi-error: statement

## Theorem 2 (contraction of the quasi-error)

Let  $0 < \theta \leq 1$ , Dörfler marking with parameter  $\theta$ , Refine bisecting every  $T \in \mathcal{M}_k$ . There are  $\gamma > 0$ ,  $\alpha \in (0, 1)$  (shape regularity, data,  $\theta$  only) with

$$\zeta_{k+1}^2 = e_{k+1}^2 + \gamma \mathcal{E}_{k+1}^2 \leq \alpha^2 \zeta_k^2 = \alpha^2 (e_k^2 + \gamma \mathcal{E}_k^2).$$

## Consequence

$$\zeta_k \leq \alpha^k \zeta_0, \quad \rightsquigarrow \quad \text{linear convergence of the quasi-error.}$$

Hence,

$$e_k \leq \zeta_k \leq \alpha^k \zeta_0 \quad \rightsquigarrow \quad \text{AFEM converges linearly, unconditionally.}$$

## Proof — Steps 1–2: assemble and choose $\delta, \gamma$

- **Pythagoras:** 
$$e_{k+1}^2 = e_k^2 - E_k^2.$$

- **Estimator reduction and Dörfler marking:** (any  $\delta > 0$ )

$$\mathcal{E}_{k+1}^2 \leq (1 + \delta)(1 - \lambda\theta^2)\mathcal{E}_k^2 + (1 + \delta^{-1})C_0 E_k^2.$$

- **Combine:** Multiply the second by  $\gamma$  and add the first:

$$\zeta_{k+1}^2 \leq e_k^2 + (\gamma(1 + \delta^{-1})C_0 - 1)E_k^2 + \gamma(1 + \delta)(1 - \lambda\theta^2)\mathcal{E}_k^2.$$

- **Choose  $\delta$ :** Fix  $\delta$  by  $(1 + \delta)(1 - \lambda\theta^2) = 1 - \frac{1}{2}\lambda\theta^2$ , i.e.  $\delta = \frac{\lambda\theta^2}{2(1 - \lambda\theta^2)} > 0$ .

- **Choose  $\gamma$ :** then  $\gamma > 0$  by  $\gamma(1 + \delta^{-1})C_0 = 1$  — this **exactly cancels** the  $E_k^2$  term:

$$\zeta_{k+1}^2 \leq e_k^2 + \gamma(1 - \frac{1}{2}\lambda\theta^2)\mathcal{E}_k^2.$$

## Proof — Step 3: Reliability

So far,

$$\zeta_{k+1}^2 \leq e_k^2 + \gamma \left(1 - \frac{\lambda\theta^2}{2}\right) \mathcal{E}_k^2.$$

● **Reliability:**  $e_k^2 \leq C_1 \mathcal{E}_k^2$ :

$$\begin{aligned} \zeta_{k+1}^2 &\leq \left(1 - \frac{\gamma\lambda\theta^2}{4C_1}\right) e_k^2 + \frac{\gamma\lambda\theta^2}{4C_1} e_k^2 + \gamma \left(1 - \frac{1}{2}\lambda\theta^2\right) \mathcal{E}_k^2 \\ &\leq \left(1 - \frac{\gamma\lambda\theta^2}{4C_1}\right) e_k^2 + \frac{\gamma\lambda\theta^2}{4} \mathcal{E}_k^2 + \gamma \left(1 - \frac{1}{2}\lambda\theta^2\right) \mathcal{E}_k^2 \\ &\leq \left(1 - \frac{\gamma\lambda\theta^2}{4C_1}\right) e_k^2 + \left(1 - \frac{\lambda\theta^2}{4}\right) \gamma \mathcal{E}_k^2 \leq \alpha^2 (e_k^2 + \gamma \mathcal{E}_k^2) \zeta_k^2 = \alpha^2 \zeta_k^2 \end{aligned}$$

with  $\alpha^2 := \max \left\{ 1 - \frac{\gamma\lambda\theta^2}{4C_1}, 1 - \frac{\lambda\theta^2}{4} \right\} < 1$ .

□

Synthesis

# Ingredients of the two proofs

- **Vanishing oscillation:** contracts  $e_k$  *itself*, but needs  $\text{osc}_k(U_k) = 0$  *and* the interior node property.
- **General case:** contracts the quasi-error  $\zeta_k^2 = e_k^2 + \gamma \mathcal{E}_k^2$ , needs *neither*.

## The four ingredients of the general proof

- Dörfler marking.
- Pythagoras orthogonality.
- The a posteriori **upper** bound (reliability).
- Estimator reduction.

Notably *not* the lower bound (efficiency) — that returns only in the convergence-*rate* analysis ahead.

## Quasi-error vs. total error

- **Total error:** Reliability and efficiency together give the *total error* equivalence

$$\mathcal{E}_k^2 \simeq e_k^2 + \text{osc}_k^2 =: \text{total error.}$$

### Two equivalent quantities, two different jobs

The quasi-error  $\zeta_k^2 = e_k^2 + \gamma \mathcal{E}_k^2$  is *equivalent* to the total error, and it is the total error that governs the optimal decay *rates* ahead. So efficiency, unused here, is exactly what ties the contracted quantity to approximation classes.

Mark is driven by  $\mathcal{E}_k$  *alone*: a single Dörfler marking suffices for contraction between consecutive steps.

## Four Five basic properties

① **Upper bound** (reliability, Lecture 5):  $e_k^2 \leq C_1 \mathcal{E}_k^2$ .

② **Lower bound** (efficiency, Lecture 5):  $C_2 \mathcal{E}_k^2 \leq e_k^2 + \text{osc}_k^2$ .

③ **Discrete lower bound** (thanks to interior node property, Lecture 5):

$$C_2 \mathcal{E}_k^2(U_k, T) \leq \|U_{k+1} - U_k\|_{\omega_k(T)}^2 + \text{osc}_k^2(U_k, \omega_k(T)).$$

④ **Pythagoras orthogonality**: Since  $\mathcal{B}[u - U_{k+1}, U_{k+1} - U_k] = 0$  ( $U_{k+1} - U_k \in \mathbb{V}_{k+1}$ )

$$e_k^2 = e_{k+1}^2 + E_k^2.$$

⑤ **Localized upper bound**  $\mathcal{R}_k^+$ : discrete neighborhood of the refined set  $\mathcal{R}_k = \mathcal{I}_k \setminus \mathcal{I}_{k+1}$

$$E_k^2 \leq C_1 \mathcal{E}_k^2(U_k, \mathcal{R}_k^+).$$

Localized upper bound



## Localized upper bound: statement

- **No local bound continuous-to-discrete:** Between the *continuous* and discrete solution, no local upper bound is possible. Between **two discrete** solutions on nested meshes, it is:

### Lemma 2 (localized upper bound)

Let  $\mathcal{T}_{k+1} \geq \mathcal{T}_k$  with refined set  $\mathcal{R}_k = \mathcal{T}_k \setminus \mathcal{T}_{k+1}$ , and  $\mathcal{R}_k^+$  its discrete  $\mathcal{T}_k$ -neighborhood. Then

$$E_k = \|U_{k+1} - U_k\|_{\Omega} \leq \sqrt{C_1} \mathcal{E}_k(U_k, \mathcal{R}_k^+), \quad \mathcal{E}_k(U_k, \mathcal{R}_k^+) = \left( \sum_{T \in \mathcal{R}_k^+} \mathcal{E}_k^2(U_k, T) \right)^{1/2}.$$

- **Interpretation:** Only the elements *actually refined* (and their neighbors) control how much the solution can change — unrefined regions contribute nothing to  $E_k$ .

## Localized upper bound: proof

- Setup:** Set  $V = U_{k+1} - U_k \in \mathbb{V}_{k+1}$ , and let  $I_k V \in \mathbb{V}_k$  be its quasi-interpolant onto the coarse space. Galerkin orthogonality on both meshes, with residual  $\langle R_k, w \rangle = \langle f, w \rangle - \mathcal{B}[U_k, w]$ :

$$\|U_{k+1} - U_k\|_{\Omega}^2 = \mathcal{B}[U_{k+1} - U_k, V] = \langle f, V \rangle - \mathcal{B}[U_k, V] = \langle R_k, V \rangle = \langle R_k, V - I_k V \rangle,$$

the last step because  $\langle R_k, I_k V \rangle = 0$ .

### The key localization

$I_k V$  reproduces  $V$  exactly wherever refinement has not touched the discrete neighborhood, so  $V - I_k V$  is supported only on  $\bigcup_{T \in \mathcal{R}_k} \omega_{\mathcal{T}_k}(T) = \bigcup_{T \in \mathcal{R}_k^+} T$ .

- Localized reliability:** Repeating the reliability argument of Lecture 5, but summing only over  $\mathcal{R}_k^+$ :

$$\langle R_k, V - I_k V \rangle = \sum_{T \in \mathcal{R}_k^+} \int_T r(V - I_k V) + \int_{\partial T} j(V - I_k V) \leq C_1 \mathcal{E}_k(U_k, \mathcal{R}_k^+) \|V\|_{\Omega}.$$

# Towards optimality: Four basic properties

- ① **Upper bound** (reliability, Lecture 5):  $e_k^2 \leq C_1 \mathcal{E}_k^2$ .
- ② **Lower bound** (efficiency, Lecture 5):  $C_2 \mathcal{E}_k^2 \leq e_k^2 + \text{osc}_k^2$ .
- ③ **Pythagoras orthogonality:** Since  $\mathcal{B}[u - U_{k+1}, U_{k+1} - U_k] = 0$  ( $U_{k+1} - U_k \in \mathbb{V}_{k+1}$ )

$$e_k^2 = e_{k+1}^2 + E_k^2.$$

- ④ **Localized upper bound**  $\mathcal{R}_k^+$ : discrete neighborhood of the refined set  $\mathcal{R}_k = \mathcal{T}_k \setminus \mathcal{T}_{k+1}$

$$E_k^2 \leq C_1 \mathcal{E}_k^2(U_k, \mathcal{R}_k^+).$$

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*Next: Lecture 8 — quasi-optimal convergence rates.*